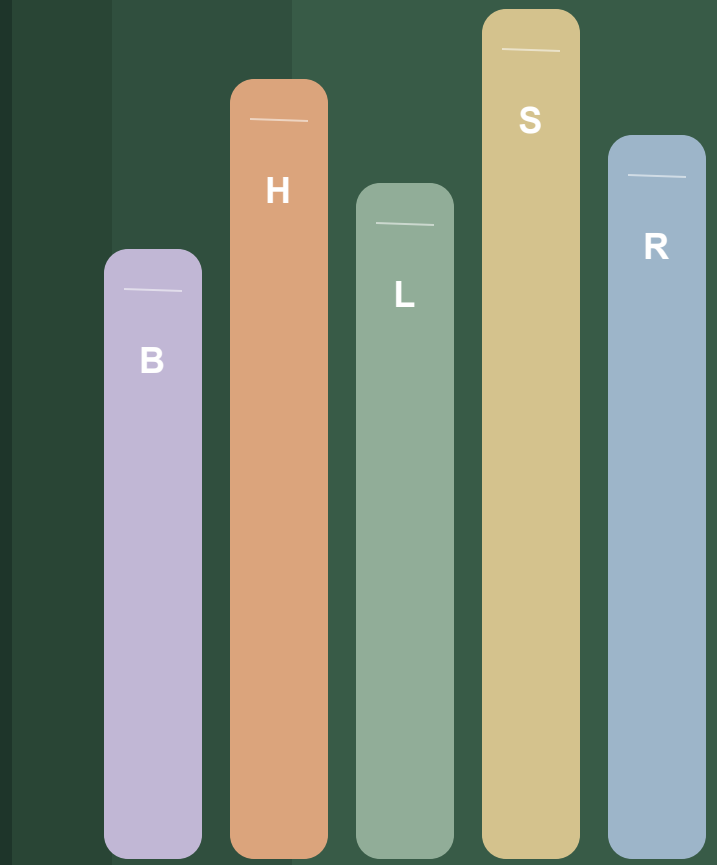
BOOK HAVEN

Library Management System

Java web application · Review 1

Catalog · Members · Loans

TEAM: ADD NAMES



Problem and solution

Library staff need one dependable view of the collection and who has borrowed each copy.

01

Finding a title

Paper lists slow down searches and make availability hard to confirm.

02

Tracking a loan

Borrower names and due dates can be missed when records are split.

03

Welcoming readers

Member details and borrowing history need a consistent home.

BOOK HAVEN

Search the catalog, register members, issue available copies, and record returns.

Library desk workflows

A short, repeatable circulation flow keeps availability and loan history aligned.

01

Search

Find by title, author, or category.

02

Select

Choose an available copy and member.

03

Issue

Save the loan and 14-day due date.

04

Return

Close the loan and restore availability.

GUARDRAIL

An issue is blocked when no copy is available; returns can only be recorded once.

Application architecture

Each layer has one job, so interface handling, business rules, and SQL stay easy to follow.



BACKGROUND TASK

Overdue scan

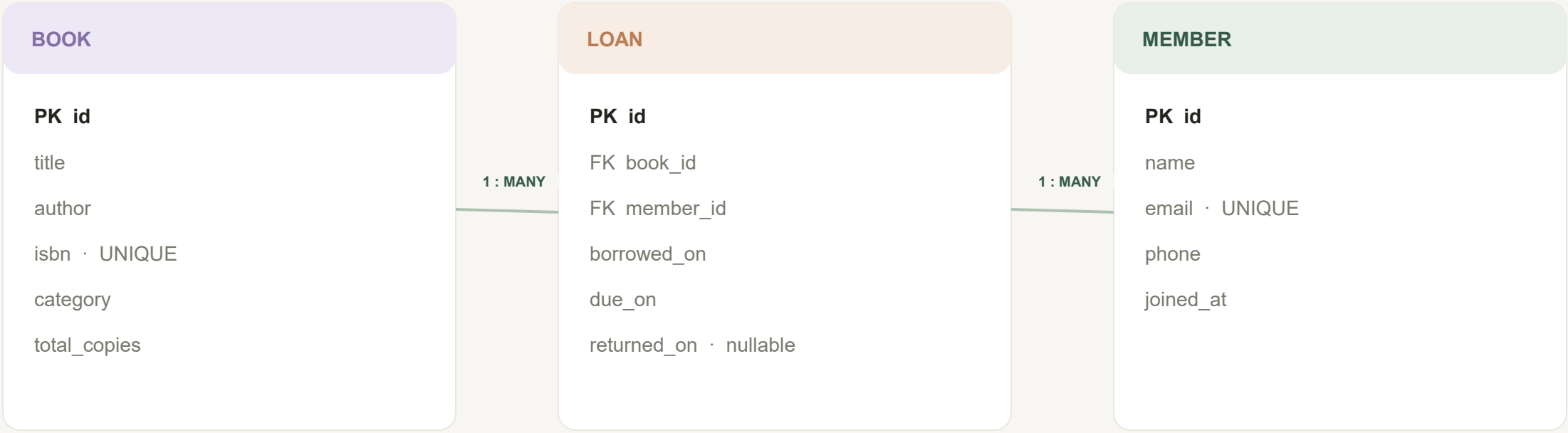
A scheduled executor checks active loans and logs overdue items.

REQUEST SAFETY

Issue transaction locks the book row while availability is checked and updated.

Database design

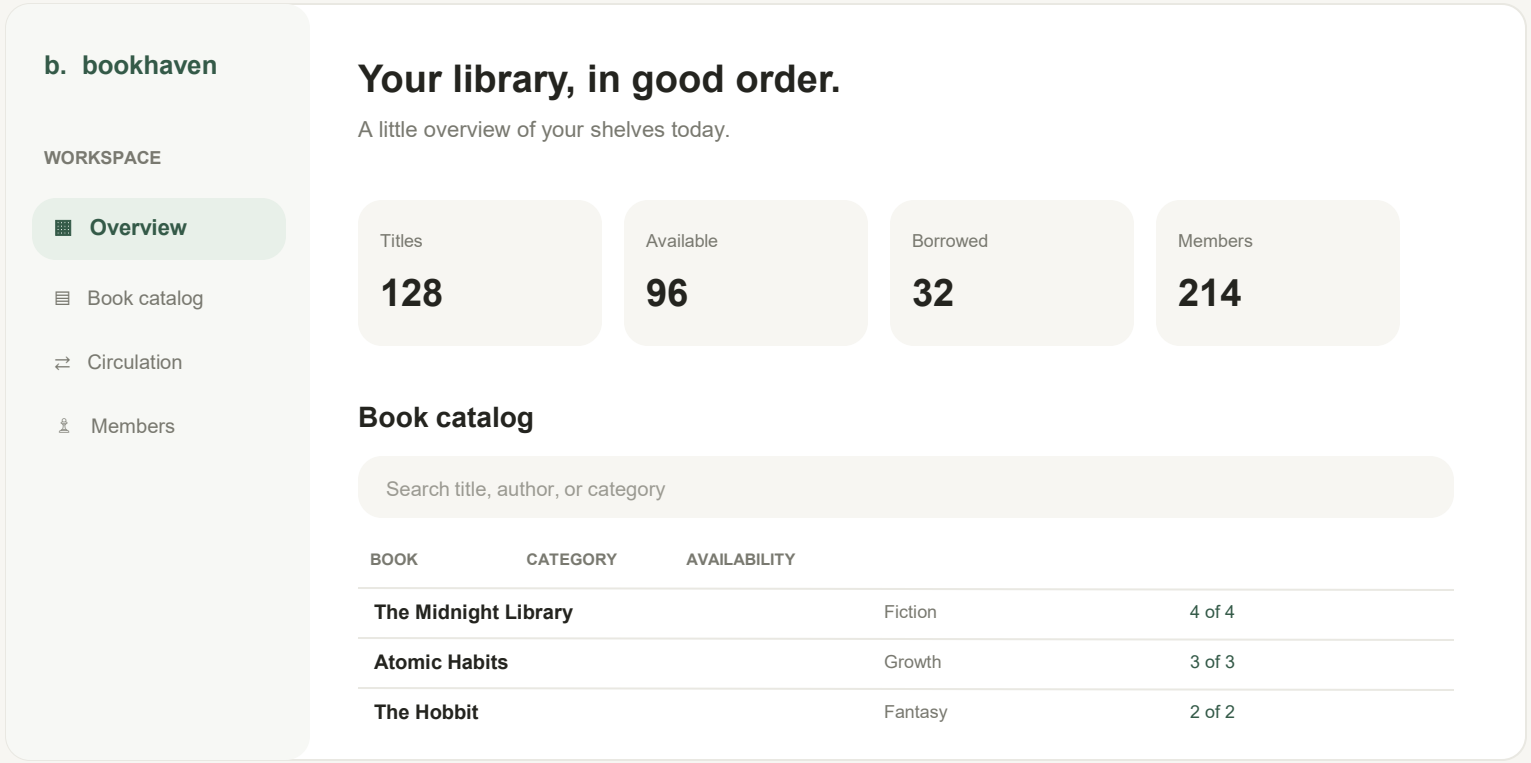
Three normalized tables keep book and member details in one place and preserve circulation history.



Availability — **total copies − active loans (returned_on is empty)**

Interface and responsive layout

The desk dashboard brings the next common action close to the information staff need.



DESIGN CHOICES

Responsive

Cards and forms reflow for smaller screens.

Clear state

Availability, overdue status, and errors are visible.

Focused actions

Add, register, issue, and return from one page.

Interface wireframe based on the implemented dashboard; sample counts above are illustrative.

Java concepts and JDBC

01	**OOP + polymorphism**	Abstract LibraryEntity with Book, Member, and Loan subclasses; shared display behavior.
02	**Interfaces + generics**	Repository contracts define typed operations and List<T> results.
03	**Exceptions**	Validation, not-found, duplicate, and database failures return clear API errors.
04	**JDBC + database classes**	JdbcBookRepository, JdbcMemberRepository, and JdbcLoanRepository use prepared statements.
05	**Transaction safety**	Issue checks availability and creates the loan in one transaction with a row lock.
06	**Concurrency**	A scheduled executor performs the overdue scan while Tomcat serves HTTP requests.

Demo and next steps

A three-minute demo can show the complete circulation loop.

1	**Search**	Find a book by title or author.
2	**Register**	Add a reader with contact details.
3	**Issue**	Choose a book and member; show due date.
4	**Return**	Mark the loan returned and check availability.

NEXT ITERATION

- Staff authentication and roles
- Fines and reminder notifications
- Deployment and production database

SETUP

JDK 21 + Maven 3.9 · mvn clean compile exec:java · http://localhost:8080